Absolute flags and their moving lines

Human Compression project: L017 working recovery

This author manuscript proposes a corrected recovery of results of Dimitri Leemans, Klara Stokes and Philippe Tranchida, *On trialities and their absolute geometries*, 21 March 2024. The geometries and claimed parameters belong to those authors. The coordinate argument below replaces their isolated small twisted-field computation. Its complete source contract and finite certificates are supplied separately in the same packet. Distinct review and whole-original acceptance remain pending. No numerical factor or formalization is claimed.

1 Classical input and conventions

Let F be a finite field, σ an automorphism of order one or three, $K = \mathrm{Fix}(\sigma)$, $f = |F|$ and $k = |K|$. In $\mathrm{PG}(7, F)$ put

$$Q(X) = X_0X_4 + X_1X_5 + X_2X_6 + X_3X_7.$$

The D_4 geometry has singular points, the two families of maximal singular three-spaces, and singular lines. Opposite-family three-spaces are incident when their intersection is a plane. All other incidence is containment.

We retain the following precise classical input from Tits's triality theory, as cited in source Section 2.3 and reference [17], and the generalized-polygon facts cited there from [18]. A triality of type I_σ is conjugate to the canonical cyclic triality followed by coordinatewise σ. Its fixed geometry is a Moufang generalized hexagon of order (f, k): each absolute line has $f + 1$ points and each absolute point lies on $k + 1$ absolute lines. The commuting projectivity group G is $G_2(k)$ or ${}^3D_4(k)$ respectively. It is transitive on apartments with a distinguished chamber, and on ordered point paths (A, P, B) of length two in the point-collinearity graph, with the two lines distinct. Any two points lie in an apartment. These classification, embedding and transitivity assertions are retained inputs, not newly proved Lie-theoretic results.

The incidence graph of this hexagon has girth twelve and diameter six. An apartment therefore has six points and six lines. When we say that two points are at distance j in the hexagon, we mean point-collinearity distance; their incidence-graph distance is $2j$. All distances in the moving geometry will instead be incidence-graph distances. This distinguishes the two conventions used in the source's counting proof.

Here is the explicit canonical coordinate model needed from the classical input. In three copies of the point coordinates set

$$\begin{aligned} T(X,Y,Z) = {} & \det\begin{pmatrix} X_0 & X_1 & X_2 \\ Y_0 & Y_1 & Y_2 \\ Z_0 & Z_1 & Z_2 \end{pmatrix} + \det\begin{pmatrix} X_4 & X_5 & X_6 \\ Y_4 & Y_5 & Y_6 \\ Z_4 & Z_5 & Z_6 \end{pmatrix} \\ & + X_3(Z_0Y_4 + Z_1Y_5 + Z_2Y_6) + X_7(Y_0Z_4 + Y_1Z_5 + Y_2Z_6) \\ & + Y_3(X_0Z_4 + X_1Z_5 + X_2Z_6) + Y_7(Z_0X_4 + Z_1X_5 + Z_2X_6) \\ & + Z_3(Y_0X_4 + Y_1X_5 + Y_2X_6) + Z_7(X_0Y_4 + X_1Y_5 + X_2Y_6) \\ & - X_3Y_3Z_3 - X_7Y_7Z_7. \end{aligned}$$

A point X is incident with the type-one object represented by Y when $T(X, Y, Z) = 0$ for every Z; the other incidences follow cyclically. The canonical triality cycles the three coordinate copies.

Its semilinear version applies σ as well. In particular an absolute point satisfies $T(X, X^\sigma, Z) = 0$ for every Z. Applying the triality supplies the other two required incidences.

2 Special planes and a uniform distant pair

Write e_i for the coordinate points. At $P = e_0$, direct comparison of coefficients in the two incidence equations gives

$$P^\alpha = \langle e_0, e_3, e_5, e_6\rangle, \qquad P^{\alpha^2} = \langle e_0, e_5, e_6, e_7\rangle, \qquad P^\omega = \langle e_0, e_5, e_6\rangle.$$

For $X = ae_0 + be_5 + ce_6$ its absolute-point condition is precisely $bc^\sigma - cb^\sigma = 0$. Thus the absolute points of this plane are the union of the $k + 1$ pencil lines through P with $[b : c] \in \mathrm{PG}(1, K)$. Each pencil line has all its $f + 1$ points absolute. These are exactly the absolute lines through P, by the retained valency. Point transitivity transfers this description to every P^ω.

The following coordinate apartment is defined over K:

$$e_0, e_5, e_2, e_4, e_1, e_6, e_0. \tag{1}$$

For completeness, in the untwisted hyperplane $X_7 = -X_3$ the fixed lines are characterized by the source's six Plücker equations

$$X_{12} = X_{34}, \quad X_{54} = X_{32}, \quad X_{20} = X_{35}, \quad X_{65} = X_{30}, \quad X_{01} = X_{36}, \quad X_{46} = X_{31}. \tag{2}$$

The six lines in (1) satisfy (2), so are fixed by the canonical triality; their K-definition makes them fixed by the semilinear triality too. The opposite pairs $(e_0, e_4), (e_5, e_1), (e_2, e_6)$ have nonzero polar pairing for Q. Apartment transitivity now proves that opposite absolute points are never joined by a singular line. This proves source Lemma 3.1 without its ambiguous subspace argument.

Lemma 1. *Every moving line contains exactly $k+1$ absolute points and lies in a unique special plane P^ω, without passing through P. Conversely the f^2 lines of P^ω not through P are moving lines.*

Proof. Take two absolute points on a moving line and put them in an apartment. They are not adjacent, since their projective join would then be fixed; they are not opposite by the preceding polar calculation. They therefore have a unique middle point P on a two-edge point path. Their join lies in P^ω and avoids P. It meets each of the $k + 1$ absolute pencil lines once, in distinct points, and these are all its absolute points by the coordinate description. Conversely any nonpencil line has those $k + 1$ absolute points and cannot be fixed: otherwise it and two pencil lines would give a triangle in the generalized hexagon. If it lay also in R^ω, any two of its absolute points would have both P and R as middle points. Girth twelve implies $P = R$. A projective plane over F has exactly f^2 nonpencil lines. □

Lemma 2. *The two moving lines $l = \langle e_5, e_6\rangle$ and $l' = \langle e_1, e_2\rangle$ have incidence-graph distance at least six, for both $f = k$ and $f = k^3$.*

Proof. They are nonpencil lines in e_0^ω and e_4^ω. Their absolute points are exactly their K-rational points: these already give $k+1$ absolute points, the full number from Lemma 1. Let $A = ae_5+be_6$ and $B = ce_1 + de_2$ be such points. Their join is singular exactly when $ac + bd = 0$. All six equations (2) for this join are $0 = 0$. Thus, if singular, it is fixed by the canonical triality; being defined over K, it is also fixed by the semilinear triality. Consequently no moving line joins an absolute point of l to one of l'. The two original lines are disjoint, so there is no path of length zero, two or four between them. This includes $(k, f) = (2, 8)$ without a finite computation. □

The unnumbered source assertion about the image of a moving line also has a direct proof. For $l = \langle e_5, e_6 \rangle$ the incidence equations give $\alpha(l) = \langle e_0, e_7 \rangle$. Substituting $X = ae_0 + be_7$ in $T(X, X^\sigma, Z)$ makes the coefficient of Z_7 equal to $-bb^\sigma$, so e_0 is its only absolute point. The transitivity proved next, and the fact that G commutes with α, transfer this to every moving line. In particular α does not act on the chosen moving-line set.

3 All parameters and distances

The retained transitivity on ordered two-edge point paths maps any moving flag (A, l) to any other: choose $B \neq A$ on l and let P be their middle point. The path (A, P, B) determines the flag. It also shows that G_P is transitive on all f^2 nonpencil lines in P^ω, recovering source Lemma 3.12.

The number of points at point distance two from an absolute point is $k(k+1)f^2$, by the unique paths of length four in the hexagon. Each moving line through that point accounts for exactly k of them, so its moving valency is

$$C = (k+1)f^2. \tag{3}$$

Counting successive levels of a generalized hexagon of order (f, k) gives

$$N = (f+1)(1 + fk + f^2k^2) \tag{4}$$

points. One can check the last level directly: the first two nonzero point levels have sizes $f(k+1)$ and $f^2k(k+1)$; the last has f^3k^2, since each last-level point is reached through all its $k+1$ lines. Summing gives (4). Double-counting flags with (3) yields Nf^2 moving lines. The elementary breadth-first bound from a moving line is exactly

$$1 + (k+1)(C-1) + k(k+1)(C-1)^2 \tag{5}$$

lines within distance four. Repeated vertices can only lower this bound. This recovers source Lemma 3.8, although Lemma 2 makes its comparison with the total unnecessary for the diameter proof.

Theorem 3. *The moving geometry is flag-transitive, has N points and Nf^2 lines, line size $k+1$ and point valency $(k+1)f^2$. Its incidence girth is six, point eccentricity is five, and line eccentricity is six.*

Proof. The geometry has no isolated elements by Lemma 1 and (3). Distinct projective lines have at most one point in common, so its incidence girth is at least six. Alternate points of the apartment (1) form a noncollinear triangle of moving joins, giving equality.

Every two absolute points have moving incidence distance at most four. For point distance two in the hexagon their moving join suffices. For adjacent points A, B, choose a third point P on their absolute line and an absolute point D on another absolute line through P. Both DA, DB are nonpencil lines of P^ω, hence moving. For opposite points, label an apartment cyclically by A, P_1, P_2, B, P_4, P_5. Choose a third point D on the line P_1P_2. Then AD lies in P_1^ω and BD in P_2^ω; both are nonpencil, hence moving.

Every point is consequently within five of every moving line, and all pairs of moving lines are within six. The latter bound is attained by Lemma 2. To attain the former, take a point A of an apartment and the moving join of its neighbor P_1 with its opposite point B. The quadric's all-or-one property, or the polar linear equation on this line, says its only point singular-collinear with A is P_1, because A is not singular-collinear with B. But AP_1 is fixed. There is therefore no moving path of length three from A to that moving line. The distance is five. Point and line transitivity give the asserted eccentricities at every element. □

The subgroup implementation following source Lemma 3.12 requires an actual geometric identification. A stabilizer of a moving line in P^ω has index f^2 in G_P, but an abstract subgroup of that index is not thereby identified with the stabilizer. A correct implementation constructs the line and its stabilizer, or verifies its orbit and incidence in the coordinate geometry. Orders and indices alone are insufficient.

4 Completing the even projective-linear construction

This section supplies the characteristic-two counterpart of the paid odd-characteristic signed-vector theorem in the accepted L023 recovery. It uses no finite computation.

Let F have characteristic two and let $a, c, b \in \mathrm{SL}_2(F)$ be involutions, with a, c distinct and commuting, such that b does not commute with any of a, c, ac. Suppose $\langle a, b, c\rangle = \mathrm{PSL}_2(F)$. Distinct commuting nonidentity unipotents have the same fixed point on $\mathrm{PG}(1, F)$; this follows directly from the centralizer of $\left(\begin{smallmatrix}1 & x\\ 0 & 1\end{smallmatrix}\right)$. Conjugate their fixed point to infinity. The matrix b cannot fix infinity, since then the generated group would be triangular. An upper unitriangular conjugation makes its diagonal entries zero, and a diagonal conjugation makes its lower-left entry one, using the unique square root in F. We therefore have

$$a = u(x), \quad c = u(y), \quad ac = u(x+y), \quad u(z) = \begin{pmatrix}1 & z\\ 0 & 1\end{pmatrix}, \qquad b = J = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}, \tag{6}$$

where $x, y, x+y$ are nonzero and distinct. Put $x_i = x, x_j = y, x_k = x+y$ and $H_i = \langle u(x_i), J\rangle$, and similarly for j, k. Both generators of H_i preserve the normalized quadratic form

$$F_i(s,t) = a_i(s^2 + t^2) + st, \qquad a_i = x_i^{-1} = t_i^2.$$

All normalized binary quadratic forms have unique coordinates $F_{u,v} = u^2 s^2 + st + v^2 t^2$. The action of $\mathrm{SL}_2(F)$ on them is faithful: preserving all their differences forces the action on every squared linear form to be trivial, hence the original linear action is trivial.

Lemma 4. *The simultaneous stabilizer of F_i, F_j is $\{1, J\}$. Consequently $H_i \cap H_j = H_i \cap H_k = H_j \cap H_k = \langle J\rangle$.*

Proof. Their difference is $(a_i + a_j)(s+t)^2$, with nonzero coefficient. A simultaneous stabilizer therefore fixes the linear form $s+t$. A determinant-one matrix doing so has the form $\left(\begin{smallmatrix}p & 1+p\\ 1+p & p\end{smallmatrix}\right)$. Preserving F_i imposes $p^2 + p = 0$, so it is I or J. These two matrices lie in both parabolics; the same proof works for every pair. □

Lemma 5. *The orbit of F_j under H_i and its orbit under H_k intersect in at most three quadratic forms.*

Proof. For an isometry of F_i, the dual action on linear forms preserves the dual form. Its coefficients are the same as those of F_i, since the polar matrix interchanges the two coordinates. Also $F_j - F_i = (t_j + t_i)^2(s+t)^2$, and $F_i(1,1) = 1$. It follows that any $F_{u,v}$ in the first orbit satisfies

$$a_i(u+v)^2 + uv + t_i(u+v) = a_j. \tag{7}$$

Explicitly apply F_i to $(u + t_i, v + t_i)$ and set its value equal to $(t_j + t_i)^2$; expansion gives (7). The second orbit satisfies the same equation with i replaced by k. Subtraction, with $w = u + v$, gives

$$(t_i + t_k)^2 w^2 + (t_i + t_k)w = 0.$$

Since $t_i \neq t_k$, either $w = 0$ or $w = (t_i + t_k)^{-1}$. The first choice gives $u = v = t_j$ uniquely. In the second case, substitute $v = u + w$ in (7); the resulting polynomial in u has degree two and leading coefficient one. It has at most two roots. We need only this upper bound, not a supposition that both roots are present. □

Consider the coset pregeometry with parabolics H_i, V, H_j, H_k, where $V = \langle a, c\rangle$. Their intersections with V are respectively $\langle a\rangle, \langle c\rangle, \langle ac\rangle$: for example $u(y)$ preserves F_i only if $a_i y^2 + y = 0$, namely $y = 0$ or $y = x_i$. In the residue of H_i, H_k, a vertex coset gH_j is sent to the form $F_j^{g^{-1}}$, with action convention chosen so that left cosets give a well-defined form. Represent g in H_i to see injectivity on this face-star from Lemma 4. Represent it also in H_k to apply Lemma 5. Thus there are at most three vertices. The three cosets

$$H_j, \quad aH_j, \quad baH_j$$

are present and distinct, just as in the odd signed-vector argument: the necessary intersections use ac and bac, and equality of the last two would force $aba = b$. There are exactly two incident edges, V, bV. Indeed $V = \langle a\rangle\langle ac\rangle$ permits its two factors to be absorbed into representatives in H_i, H_k; their intersection is $\{1, b\}$. These edges join the first vertex to each of the other two.

Theorem 6. *The characteristic-two pregeometry is a geometry. Each face–Petrie residue is the path with three vertices and two edges; it has two free group orbits of chambers and is not firm.*

Proof. The preceding calculation completes every flag of the three nonedge types. Every flag containing an edge completes because its three other types form a complete three-partite structure, with two choices in each part. Incident pairs avoiding edges have common representatives and also extend. These are all flags. At an edge the eight chambers split into two free V-orbits, and hence globally into two free group orbits. The two endpoints of the displayed residue each have just one completing edge, proving nonfirmness. The exact pair intersections also prove the usual faithful thin rank-three map conditions; they are not inferred from an arbitrary quotient of the universal map group. □

5 The fixed absolute of a cubic projective-linear marking

Let $G = \mathrm{PSL}_2(q^3)$ and let a, b, c satisfy the five hypotheses of source Section 3.2: they are generating involutions, a, c commute, an actual order-three outer automorphism α cycles a, c, ac and fixes b, and no marked duality exists. Writing $\alpha \in \mathrm{Out}(G)$ in the source must mean an actual automorphism representing that outer class; otherwise its action on elements and cosets is undefined. The three periods $|ab|, |cb|, |(ac)b|$ are equal and exceed two. If they were at most two, all three generators would commute, contrary to generation of this nonabelian simple group.

The odd-characteristic completion theorem retained in full from the accepted L023 packet, and Theorem 6 in characteristic two, apply. In particular all flags complete, all three outer-parabolic pair intersections are $\langle b\rangle$, and the exact pair residue is a path with three vertices and two edges. The rank-three truncation is the faithful map obtained by gluing its flag triangles: all its panels are thin, its flag-adjacency graph is connected by generation, and its vertex and face residues are the marked dihedral cycles. Unique flag adjacencies imply that a map automorphism fixing a flag fixes every flag. The regular G-action is therefore its full map automorphism group.

This also justifies the source's triality-without-duality assertion for these particular four-type geometries. The edge type has $|G|/4$ objects, whereas every other type has $|G|/(2m)$, with $m > 2$ the common period. Any correlation fixes the edge type. If it induced a transposition on the remaining three types, conjugating by the known triality would give a vertex–face interchange fixing the Petrie type. Its restriction to the faithful map would be a duality. Compose with an element of the flag-regular group to place its image base flag back at the base flag; the induced marked automorphism of G then interchanges a, c and fixes b, contrary to source hypothesis (5). Thus the exact type-permutation image is C_3.

We retain the standard projective-linear descent fact used in the source and the accepted cubic-generation proof: an order-three outer automorphism of $\mathrm{PSL}_2(q^3)$ is conjugate in $\mathrm{PGL}_2(q^3)$ to one of the two generating field automorphisms, and its fixed group is isomorphic to $\mathrm{PSL}_2(q)$.

This uses the odd degree of the field extension and is not a consequence merely of the word "outer."

Theorem 7. *The fixed absolute, whose objects are minimal invariant flags for the two type orbits, is a disjoint union of* $|\operatorname{PSL}_2(q)|/2$ *paths of length two. It has* $|\operatorname{PSL}_2(q)|$ *fixed edges and* $3|\operatorname{PSL}_2(q)|/2$ *absolute outer objects.*

Proof. Use left cosets, with $H_i = \langle a, b\rangle$, $H_j = \langle c, b\rangle$, $H_k = \langle ac, b\rangle$ and $V = \langle a, c\rangle$. Every stable coset of V has exactly one α-fixed element. Indeed its four elements split into orbits of lengths one or three. Four fixed elements would make all their quotients in V fixed, whereas α cycles its three nonidentity elements. Conversely a fixed element gives a stable coset. Thus fixed edges are in bijection with $C_G(\alpha)$.

At the fixed edge V, the triality cycles three two-element outer-type sets, so there are exactly two invariant outer triples. One is (H_i, H_j, H_k), whose common edges are V, bV by the exact pair residue. Call such a triple a center. The other is

$$(cH_i, acH_j, aH_k). \tag{8}$$

It has no common group representative. In fact the intersection of the first two cosets is $\{ac, acb\}$, and neither lies in aH_k, since $c \notin H_k$ and $b \in H_k$. Call such a triple a leaf.

Every outer triple extends to an edge and is a translate of one of these two triples, because the eight local chambers form two free V-orbits. These two classes cannot coincide: existence of a common representative is invariant under translation. The center has two completing edges. The leaf has just one. To verify the latter precisely, translate (8) to (H_i, H_j, cH_k). Its stabilizer is contained in $H_i \cap H_j = \{1, b\}$. If b stabilized the last coset, then cbc would lie in $H_k \cap H_j$ and equal b, contrary to $|bc| > 2$. Its stabilizer is therefore trivial. There are $|G|$ leaf triples and $|G|$ chambers in their chamber orbit; each leaf consequently has one completing edge.

Any invariant outer triple has one or two completing edges. The order-three automorphism fixes each of them, so there are no additional isolated absolute objects. Every fixed edge meets one center and one leaf, as at V; its fixed representative translates the local calculation by an element commuting with α. Centers have degree two and leaves degree one. The absolute is therefore a disjoint union of two-edge paths. Its number of edges is $|C_G(\alpha)| = |\operatorname{PSL}_2(q)|$, proving the formula. □

This proof also gives a small exact construction for the moving absolute. Let $H = C_G(\alpha)$. All fixed edges are hV with $h \in H$, and every absolute outer object is incident with one of them. Its complete candidate list is therefore the two outer cosets at each hV, with duplicates removed. For each such object, enumerate its incident original edges and keep an edge precisely when it is moved by α and its other endpoint is also absolute. This is an exhaustive construction, not a heuristic sample of the large group. Isolated vertices can then be removed using the source's stated convention.

The symmetric-group centralizer in source Step 2 cannot supply the desired triality by conjugation: every element of that centralizer induces the identity automorphism of G. Instead use the explicitly given semilinear field automorphism (or the normalizer with the action verified). Also $\operatorname{Aut}(\operatorname{PSL}_2(q^3))$ is not uniformly an extension of order three of G: the full field and diagonal automorphisms must be accounted for. The incidence and edge-fixedness tests in Steps 3–5 remain valid once this actual automorphism and the correct parabolics are supplied.